

The empirical recurrence for OEIS A250455: recovering a conjecture that is not written down

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Abstract

OEIS A250455 records “Empirical recurrence of order 83”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 84$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 83, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 83 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A250455 is “Number of $(n+2) \times (1+2) 0..2$ arrays with nondecreasing maximum of every three consecutive values in every row and column.”. It has offset 1 and begins

19683, 373248, 6859000, 120553784, 1990865512, 32248529487, 512192024001, 7998800059999, ...

The entry states:

Empirical recurrence of order 83 (see link above).

As of the “Last modified” line on the live entry (Nov 17 2022 18:07:42, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 19683$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 84$ of the 19683, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 84$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 39366$ exact terms of the model returns order 83 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{83} c_i a(n-i),$$

c_1	26	c_{22}	−47254484614085256	c_{43}	4363442881990245156342272	c_{64}	−425
c_2	−131	c_{23}	409795649803275388	c_{44}	−13467666138770649001872384	c_{65}	485
c_3	274	c_{24}	165935097582188892	c_{45}	−945397135544479214571520	c_{66}	552
c_4	−24736	c_{25}	1586311953181590308	c_{46}	−19543505874734553106501632	c_{67}	235
c_5	172860	c_{26}	−11599600828697253956	c_{47}	52767832521708712365056000	c_{68}	−232
c_6	126584	c_{27}	−3757583724298355212	c_{48}	3917044195977462567698432	c_{69}	−25
c_7	8559956	c_{28}	−38931220134980108080	c_{49}	66737046617519664138452992	c_{70}	−88
c_8	−76941044	c_{29}	241211082193769885244	c_{50}	−157218639905749321288777728	c_{71}	767
c_9	−61305572	c_{30}	59350809558897807816	c_{51}	−12987979921335143188267008	c_{72}	720
c_{10}	−1587068708	c_{31}	704573647557653723564	c_{52}	−173166496647620409394200576	c_{73}	215
c_{11}	17056287564	c_{32}	−3741478346828216245769	c_{53}	355020166347445025566097408	c_{74}	−16
c_{12}	11819426896	c_{33}	−668579648333872274442	c_{54}	32728041421976714972495872	c_{75}	−12
c_{13}	190800643380	c_{34}	−9504702603237056364531	c_{55}	339449184034397143219306496	c_{76}	−31
c_{14}	−2221250816840	c_{35}	43741456293825844036958	c_{56}	−604100289441573898398203904	c_{77}	222
c_{15}	−1364247821116	c_{36}	5551114744860145671872	c_{57}	−61147543237022349294305280	c_{78}	10
c_{16}	−16343652387486	c_{37}	96522595591485010709144	c_{58}	−498328139408298376718974976	c_{79}	24
c_{17}	185210829506864	c_{38}	−387908576540543745493744	c_{59}	767317433323139330924871680	c_{80}	−1
c_{18}	101799546823774	c_{39}	−35934135555712265082432	c_{60}	83411412573202908765487104	c_{81}	—
c_{19}	1026693769712720	c_{40}	−743482335737508021317696	c_{61}	540710426217773237786378240	c_{82}	—
c_{20}	−10415321367453888	c_{41}	2618314603515404065570944	c_{62}	−717555126005200037884198912	c_{83}	4
c_{21}	−5014476771012420	c_{42}	195963124892684440691712	c_{63}	−81409474807362013741711360		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 83, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $83 + 4$ terms removed returns order 83 as well, so $\deg q_{\min} = 83 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $83 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 83 that this sequence satisfies — and that family turns out to have exactly one member.

References

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